

MSME HACKATHON 6.0

ZERO DEFECT PROTOTYPE

InspectAI ZED Architecture

The Ultimate Edge-AI Visual Defect Detection System for Indian MSMEs!



Zero
LATENCY



100%
TRACEABILITY

STAGE 1 : CAPTURE Vision Hardware

Ultra-Affordable Optics!
Uses standard 1080p Webcams or Smartphone cameras. No expensive industrial lenses needed!

Shadow-Free Lighting
Custom LED ring-light jigs guarantee perfect, glare-free PCB illumination every single time.

USB 3.0 1080P CMOS PLUG & PLAY

STAGE 2 : PROCESS Edge-AI Engine

Lightning-Fast YOLO!
Optimized YOLOv8 object detection runs completely locally on a basic laptop. Zero cloud latency!

Absolute Precision
Instantly spots missing SMD chips, solder bridges, bent pins, and wrong polarity in milliseconds!

YOLOV8 OPENCV PYTHON

STAGE 3 : INTERFACE No-Code App

Visual Red-Boxing!
The AI forcefully draws bright RED bounding boxes exactly over the defect. Impossible to miss!

Clear Audio Alerts
Massively reduces worker fatigue with simple binary PASS (green/beep) or FAIL (red/buzzer) signals!

REACT.JS WEBSOCKETS
AUDIO API

STAGE 4 : IMPACT ZED Outcomes

Incredible ROI!
Completely eradicates costly product returns and reworks. Pays for itself in under 2 months!

100% Traceability
Automatically generates QR-coded, unalterable digital audit reports for every single batch shipped.

ZED COMPLIANT POSTGRESQL
CLOUD SYNC